

Practical Work #1

Operational amplifiers

Objectives:

- Analyze electrical networks by obtaining their input impedance
- Evaluate advantages and disadvantages of different implementations
- Analyze electrical networks by obtaining their voltage transfer
- Recognize different types of filters
- Associate circuit topologies with their frequency response

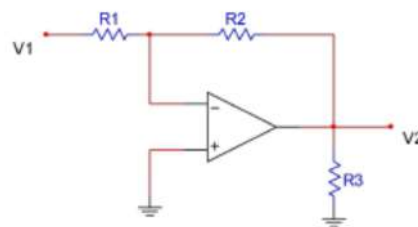
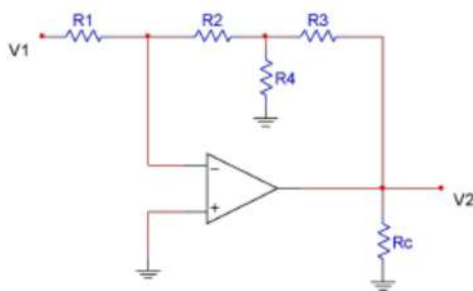
Group: groups will be a maximum of 5 people

Approval condition: the practical work must be submitted within up to 2 weeks after it is carried out in class.

Responsible: each practical work must have a person responsible. Submissions without the corresponding cover page will not be accepted.

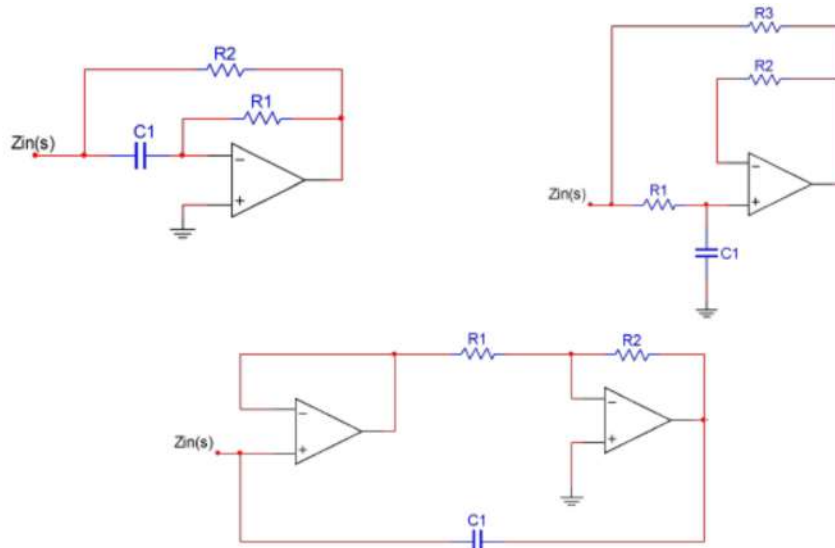
Exercise #1

Starting from the following circuit structures, it is required to design an inverting amplifier with input impedance $Z_{input}=10k\Omega$ and gain of 3000. Analyze advantages and disadvantages of each circuit. Simulate the circuit to verify the design conditions.

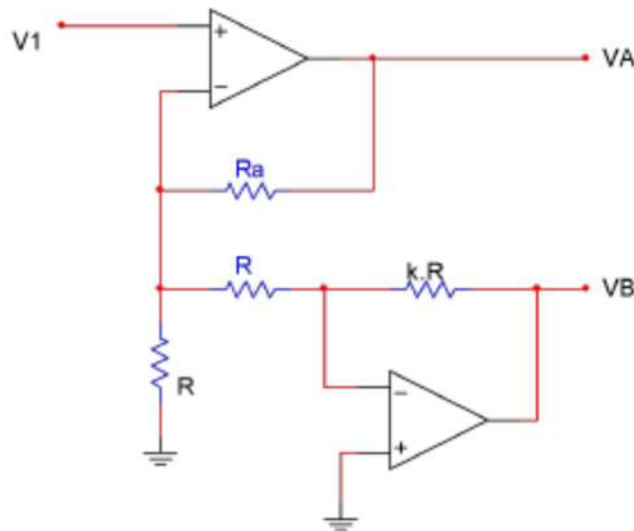


Exercise #2

Given the following dipoles, obtain the excitation impedance. Draw conclusions.



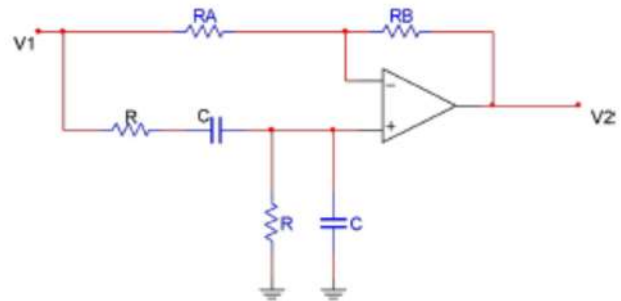
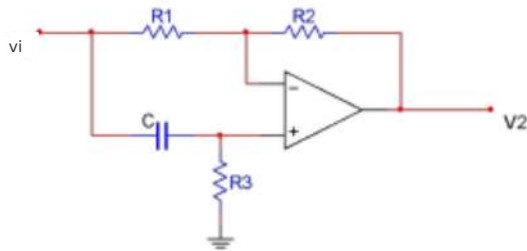
Exercise #3 Differential-output amplifier. Obtain the expression $V_A - V_B$. Use $R_a = R(k-1)/2$ as the value.



Exercise #4

For the following circuits known as All-Pass Filter or Phase Rotator, it is requested: Obtain the transfer function V_2/V_1 (magnitude, phase, and poles-and-zeros diagram) if $R_2/R_1 = 1$ and $R_A/R_B = 5$

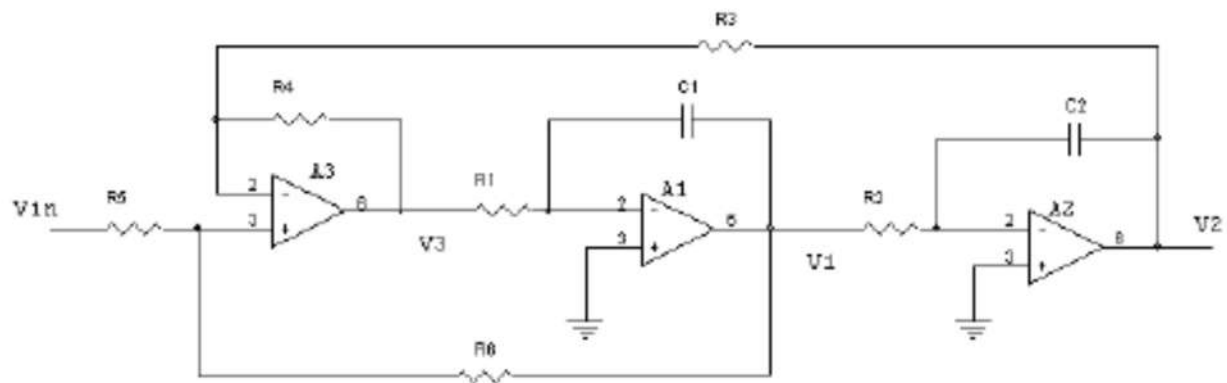
Use a simulator to obtain the phase transfer of both circuits, assuming $R=R_3=1K$; $C=1$ microF. Draw conclusions



Exercise #5

Given the following network, known as a state-variable filter, you are asked to obtain the transfers and compare them.

- V_2/V_{in}
- V_3/V_{in}
- V_1/V_{in}

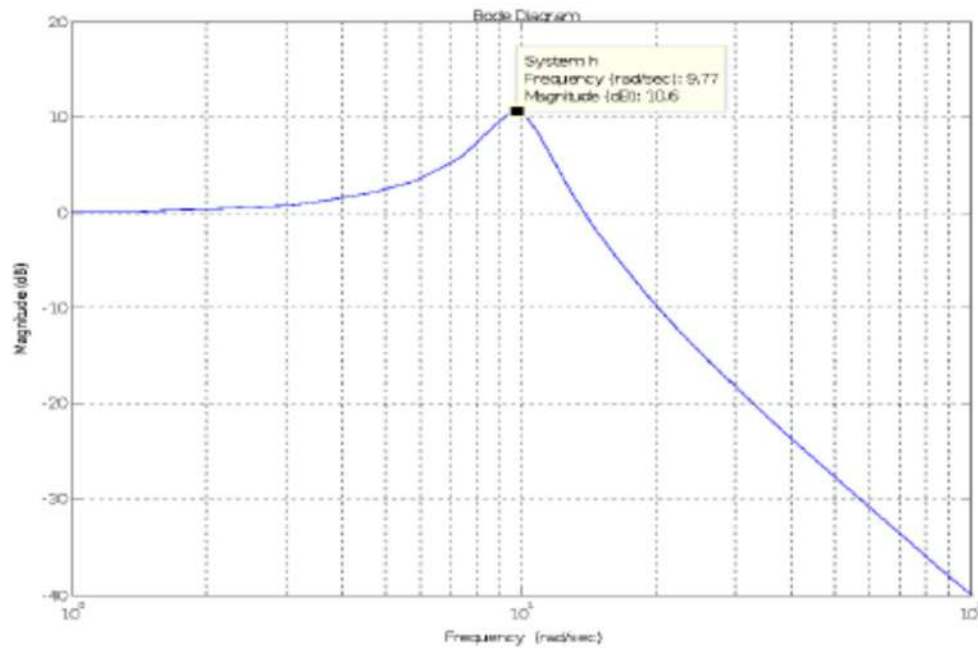
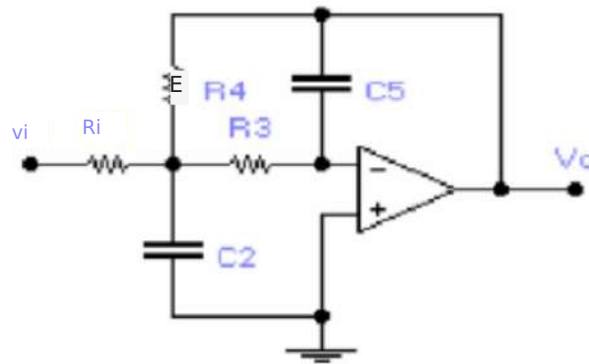


Exercise #6

Given the following circuit, the task is:

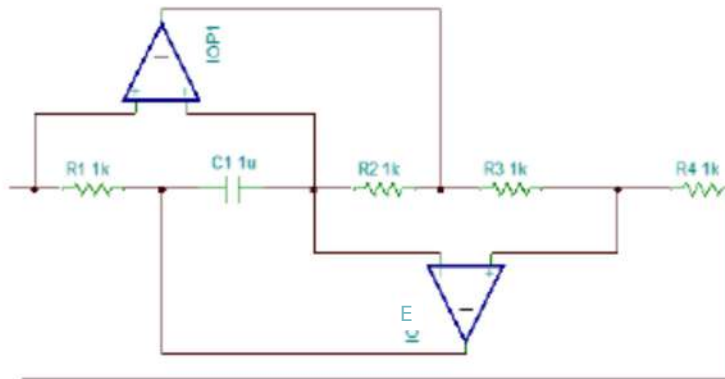
- Determine the voltage transfer of the filter, which is specified in the circuit.
- Recalculate the value of the components that make up the circuit if you want $\omega_0 = 1000$ rad/s and you have capacitors of 4700 pF and 47 pF.

Data: $R_1 = R_3 = R_4 = 1$, $C_2 = 1$, $C_5 = 0.01$



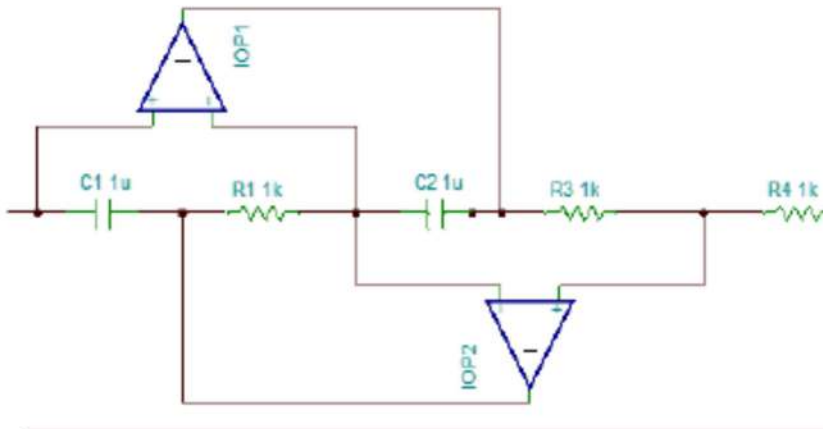
Exercise #7

To the following active dipole determine the Excitation Impedance. Graph the magnitude and phase of the Impedance interpreting the results.



Exercise#8

To the following active dipole determine the Excitation Impedance. Graph the magnitude and phase of the Impedance interpreting the results.



Exercise #9

For the following active dipole determine the Excitation Impedance. Indicate/propose applications for this network.

